Game Theory
Lecture 10+11: Knowledge

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Outline

1. (Common) Knowledge
   - The hat game
   - A model of knowledge
   - Common knowledge
   - Agree to disagree?
   - Knowledge and solution concepts
   - Rubinstein's electronic mail game

The hat game I

Example (hat game)

- n players and each one wears a hat
- hat can be either black or white
- Each player observes the other player's hat color but not his own!
- several rounds: In each round, those players who know the colour of their own hat for sure raise their hand.

Does anyone ever raise his hand?

- Say, you see: $\triangle$ $\blacktriangle$ $\triangle$
- 1st round: will you raise your hand?
- 2nd round: if no one raised hand in first round, will you raise hand? what if k players raised hand in 1st round?
- etc.

The hat game II

Example (hat game variation)

- outside observer announces truthfully before the first round: "At least one of you wears a black hat."
- If k people wear a black hat, will anyone raise his hand? Who and in which round?

Say you see: $\triangle$ $\blacktriangle$ $\triangle$

- 1st round: raise hand or not?
- 2nd round:
  - if no one raised hand in round 1: raise hand now?
  - if 1 player raised hand in round 1: raise hand now?
- who will raise hand in 3rd round?

The hat game III

Let's look at n players, $k \leq n$ of which have a black hat: For general n:

- If k=1, in which round does the black hat wearer know? In which round do the white hat wearer know?
- If k=2, which round do the black hats know? Which round the white hats?
- If k=3,. . .

The hat game IV

Result

In round k, everyone with a black hat raises his hand. In round $k+1$ everyone else raises his hand.

- take a situation with $k \geq 2$
- observer does not give any new information to any player when saying that there is at least 1 black hat
- outcome is completely different!
- observer does not create knowledge but makes sure that knowledge is *common knowledge*

A model of knowledge I

- *finite* set of states Ω
- *information function* P assigning to each $\omega \in \Omega$ a non-empty subset of Ω ($P : \omega \to 2^{\Omega}$) such that
  - (P1) $\omega \in P(\omega)$ for every $\omega \in \Omega$
  - (P2) if $\omega' \in P(\omega)$ then $P(\omega) = P(\omega')$

  i.e. information function is *partitional*

Example

Let $\Omega = [0, 1)$. The decision maker observes only the first digit of ω. In this case, the partition is $\{[0, 0.1), [0.1, 0.2), \ldots, [0.9, 1)\}$.

- an *event* is a subset of Ω, i.e. a set of states, for example
  - the interval $(0.453, 08723)$
  - the set $\{0.34, 0.56, 0.9\}$
  - hat game: "more than two people wear a black hat"
  - Ω itself is an event!

A model of knowledge II

Knowledge:

- If $P(\omega) \subseteq E$, then the decision maker *knows* that E occurred whenever the state is ω.
- the *knowledge function* is

$$K(E) = \{\omega \in \Omega : P(\omega) \subseteq E\}. \tag{1}$$

For any event E, $K(E)$ is the set of states in which the decision maker knows that E occured. (could be $\emptyset$!)

A model of knowledge III

Example (hat game in our model)

- a state is $c = (c_1, \ldots, c_n)$ where $c_i \in \{B, W\}$
- $\Omega = \{c \in \{B, W\}^n\}$
- information of player i before round 1 is $P_i^1(c) = \{(c_{-i}, W), (c_{-i}, B)\}$ (before outside announcement)
- event "i knows the color of his hat" is $E_i = \{c : P_i(c) \subseteq \{c : c_i = B\}$ or $P_i(c) \subseteq \{c : c_i = W\}\}$
- with outside observer
  - round 1 if $k = 1$ and $c_i = B$: $P_i^1(c) = \{c\}$ and $P_j^1(c) = \{(c_{-j}, W), (c_{-j}, B)\}$ for $j \neq i$
  - round 2 if $k \geq 2$: then everyone knows in round 2 that $\omega \notin F^1 = \{c : |\{i : c_i = B\}| = 1\}$; therefore, $P_i^2(c) = P_i^1(c) \backslash F^1$
    $\Rightarrow$ if $k = 2$, players with $c_i = B$ know that $c_i = B$
  - if no one raised his hand in stage m, the information is $P_i^m(c) = P_i^1(c) \backslash (F^1 \cup \cdots \cup F^{m-1})$.

Common knowledge I

For simplicity, we restrict ourselves to 2 player games.

Definition

Let K_1 and K_2 be the knowledge functions of player 1 and 2. An event $E \subseteq \Omega$ is *common knowledge* between 1 and 2 in state ω if ω is a member of every set in the sequence $K_1(E)$, $K_2(E)$, $K_1(K_2(E))$, $K_2(K_1(E))\ldots$.

- E is common knowledge if both players know and both players know that both players know and so on

Common knowledge II

Definition

Let P_1 and P_2 be the information functions of the two players. An event F is *self-evident* between 1 and 2 if for all $\omega \in F$ we have $P_i(\omega) \subseteq F$ for $i = 1, 2$.
An event E is *common knowledge* between 1 and 2 in the state ω if there exists a self-evident set F with $\omega \in F \subseteq E$.

- F is self evident if **whenever** F occurs (no matter which state $\omega \in F$) both players know that F occurred
  (and because of this both players know that both players know ...)

Common knowledge III

Example

- $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6\}$
- $P_1 = \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4, \omega_5\}, \{\omega_6\}\}$
  $P_2 = \{\{\omega_1\}, \{\omega_2, \omega_3, \omega_4\}, \{\omega_5\}, \{\omega_6\}\}$
- Questions:
  - Is the event $F = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5\}$ self evident?
  - In which states is F common knowledge?
  - Show that the event $E = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ cannot be common knowledge in any state.

Common knowledge IV

Different interpretations of a "self evident" set F

- F is self-evident between 1 and 2:
  for any $\omega \in F$ we have $P_i(\omega) \subseteq F$ for $i = 1, 2$
- this implies $F = \cup_{\omega \in F} P_1(\omega) = \cup_{\omega \in F} P_2(\omega)$
- it also means that we can write F as union of a number of disjoint partition members of each P_i
- and therefore $K_1(F) = K_2(F) = F$

Common knowledge V

Proposition

The two definitions of common knowledge are equivalent.

Proof. □

Agree to disagree? I

- players predict the probability of some event
- players start with the same prior
- players then get private information through their information function
- can it be common knowledge that players assign different probabilitites to the event?
- No!
- intuitive story:
  - each player sees other players' predictions and can revise own prediction
  - if other players' prediction is different from yours, they have some information you do not have
  - $\Rightarrow$ Bayesian updating $\Rightarrow$ revise prediction
  - repeat infinitely...
  - eventually everyone should have same prediction

Agree to disagree? II

- common prior belief is ρ (probability measure on Ω)
- $\rho(E|P_i(\omega))$ is the posterior probability player i assigns to event E in state ω
- the event "player i assigns probability η_i to event E" is the set of states $\{\omega : \rho(E|P_i(\omega) = \eta_i\}$.

Agree to disagree? III

Theorem (Aumann 1976)

Assume players have a common prior. Suppose it is common knowledge at ω' that P1's posterior probability of event E is η_1 and that P2's posterior is η_2. Then $\eta_1 = \eta_2$.

Proof. $\square$

Agree to disagree? IV

- the theorem works only if the posterior is *common knowledge*
- it is not enough that each player knows the posterior of the other:

Agree to disagree? V

Example

- $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$
- prior: each state is equally likely
- P_1 partition: $\{(\omega_1, \omega_2), (\omega_3, \omega_4)\}$
- P_2 partition: $\{(\omega_1, \omega_2, \omega_3), (\omega_4)\}$
- $E = \{\omega_1, \omega_4\}$
- take $\omega' = \omega_1$, then:
  - $\eta_1(E) = 1/2$ and $\eta_2(E) = 1/3$
  - P1 knows that P2's information is $(\omega_1, \omega_2, \omega_3) \Rightarrow$ P1 knows $\eta_2(E) = 1/3$
  - P2 knows that player 1 either knows (ω_1, ω_2) or knows (ω_3, ω_4) and in both cases $\eta_1(E) = 1/2 \Rightarrow$ P2 knows that $\eta_1(E) = 1/2$

Agree to disagree? VI

The problem in the example is that the posteriors are not *common knowledge*:

- P2 does not know what P1 thinks what $\eta_2(E)$ is:
  - for example $\omega_3 \in P_2(\omega')$ but in ω_3 P1 does not know P2's posterior
- formally: take the event F: "player 2's posterior for the event E is $1/3$"
  - F "happens" in the states ω_1, ω_2 and ω_3
  - $K_1(F) = \{\omega_1, \omega_2\}$ and $K_2(K_1(F)) = \emptyset$
  - F is not common knowledge although both players know F in state ω_1

Knowledge and solution concepts I

- strategic form game $G = \langle N, (A_i), (u_i) \rangle$
- Ω describes a set of states that describe the environment relevant to the game, i.e. $\omega \in \Omega$ consists for each player i of
  - $a_i(\omega) \in A_i$ the action chosen by player i in state ω
  - $\mu_i(\omega)$ a probability measure over A_{-i}, the belief player i holds over the other player's actions.
  - $P_i(\omega) \subseteq \Omega$ the player's knowledge (using a partitional information function)

Knowledge and solution concepts II

Proposition

Suppose G is a two player game and that in state ω each player $i \in N$

- *knows the other player's belief: $P_i(\omega) \subseteq \{\omega' : \mu_j(\omega') = \mu_j(\omega)\}$*
- *has a belief that is consistent with his knowledge: the support of $\mu_i(\omega)$ is a subset of $\{a_j(\omega') : \omega' \in P_i(\omega)\}$*
- *knows that the other player is rational: for any $\omega' \in P_i(\omega)$ the action $a_j(\omega')$ is a best reponse for player j given his belief $\mu_j(\omega')$.*

Then the mixed strategy profile $(\mu_2(\omega), \mu_1(\omega))$ is a mixed strategy Nash equilibrium.

Proof. Let a_j^* be an action in the support of $\mu_i(\omega)$. To show: a_j^* is a best response of player j given his belief $\mu_j(\omega)$. ... $\square$

Knowledge and solution concepts III

Example (clarifying notation)

	Go 2	Stop 2
Go 1	-1,-1	4,1
Stop 1	1,4	3,3

symmetric mixed strategy NE: 1/3 and 2/3.
4 states differing in actions and information but we assume same beliefs: $\mu_1(\omega_k) = (1/3, 2/3)$ and $\mu_2(\omega_k) = (1/3, 2/3)$

- ω_1: Actions: $a_1(\omega_1) = Go$, $a_2(\omega_1) = Go$;
  Information: $P_1(\omega_1) = \{\omega_1, \omega_2\}$ and $P_2(\omega_1) = \{\omega_1, \omega_3\}$
- ω_2: Actions: $a_1(\omega_2) = Go$, $a_2(\omega_2) = Stop$;
  Information: $P_1(\omega_2) = \{\omega_1, \omega_2\}$ and $P_2(\omega_2) = \{\omega_2, \omega_4\}$
- ω_3: Actions: $a_1(\omega_3) = Stop$, $a_2(\omega_3) = Go$;
  Information: $P_1(\omega_3) = \{\omega_3, \omega_4\}$ and $P_2(\omega_3) = \{\omega_1, \omega_3\}$

Knowledge and solution concepts IV

Example (calrifying notation continued)

- ω_4: Actions: $a_1(\omega_4) = Stop$, $a_2(\omega_4) = Stop$;
  Information: $P_1(\omega_4) = \{\omega_3, \omega_4\}$ and $P_2(\omega_4) = \{\omega_2, \omega_4\}$

Let us check the requirements above:

- "knows the other's beliefs": all player hold the same belief in every state
- "belief consistent with knowledge": each $P_i(\omega_k)$ includes one state with Go_{-i} and one state $Stop_{-i}$
- "best response": given the beliefs, each player is indifferent between 2 actions
- Note that for different beliefs, the third requirement will not be satisfied!

- proposition above does not hold for games with more than 2 players (see OR 5.4 for example)

Knowledge and solution concepts V

Proposition

Let G be a 2 player game and assume that in state ω it is common knowledge that each player's belief are consistent with his knowledge and that each player is rational.
Technically, suppose that there is a self-evident F with $\omega \in F$ such that for any $\omega' \in F$ and each $i \in N$

1. *the support of $\mu_i(\omega')$ is a subset of $\{a_j(\omega'') : \omega'' \in P_i(\omega')\}$*
2. *the action $a_i(\omega')$ is a best response of i given $\mu_i(\omega')$.*

Then $a_i(\omega)$ is rationalizable in G.

Proof. (skipped) □

- informational requirements of a rationalizable strategy are weaker than those for Nash equilibrium (no know other's belief)

Rubinstein's electronic mail game I

	A	B
A	M,M	1,-L
B	-L,1	0,0

a with prob $1 - p$

	A	B
A	0,0	1,-L
B	-L,1	M,M

b with prob p

Table: Electronic mail game

- with probability $p < 1/2$ the payoffs are as in b
- with probability $1 - p$ payoffs are as in a
- assume that $1 < M < L$
- $\Rightarrow$ players would like to coordinate on playing A in game a and on B in game b
- A is a dominant strategy in a $\Rightarrow (A, A)$ is the only NE in a

Rubinstein's electronic mail game II

- game b: two pure strategy equilibria (A, A) and (B, B) and a mixed one
- Note: in b, it is risky to choose B

First Version:

- P1 knows which game is played and P2 does not
- Bayesian NE
  - in game a, P1 will always play A
  - what should P2 play?
  - what should P1 play in b?
  - the unique BNE is:

Rubinstein's electronic mail game III

Real email game

- if the game is b, P1's computer will automatically(!) send a message to P2's computer
- if the game is a: no message
- every message (by any player) is automatically (!) confirmed by the recepients computer with another message
- there is a small probability $\varepsilon > 0$ that any given message is lost and does therefore not arrive
- before taking their actions, both players see the number of messages that their computer sent away

Rubinstein's electronic mail game IV

Formalization of email game::

- State space is $\Omega = \{(q_1, q_2) : q_1 = q_2 \text{ or } q_1 = q_2 + 1,\ q_i \in \mathbb{N}\}$
- $P_1(q, q) = P_1(q, q-1) = \{(q, q), (q, q-1)\}$ for $q \geq 1$ and $P_1(0, 0) = \{(0, 0)\}$
- $P_2(q, q) = P_2(q+1, q) = \{(q, q), (q+1, q)\}$
- the game played: $G(0, 0) = a$ and b otherwise
- prior belief over Ω derived from the technology: $p_i(0, 0) = 1 - p$ (i.e. game a with prob 1-p); $p_i(q+1, q) = p\varepsilon(1-\varepsilon)^{2q}$ and $p_i(q+1, q+1) = p\varepsilon(1-\varepsilon)^{2q+1}$

Rubinstein's electronic mail game V

First analysis:

- P2 knows the game in all states apart from (0,0) and (1,0).
- if the game is b, P1 does not know that P2 knows the game in states (1,0) and (1,1). Otherwise, P1 knows that P2 knows.
- in states, (1,1) and (2,1) P2 knows the game but does not know whether P1 knows that P2 knows ...

- What would you do if you read 26 on your screen?

Rubinstein's electronic mail game VI

Proposition

The electronic mail game has a unique Nash equilibrium in which both players always choose A.

Proof. The proof is by induction.

1. In state $(0, 0)$ P1 chooses ...
2. If P2 does not receive any message, he plays ...
3. Let $\tilde{q}_i$ be the lowest number of own sent messages for which player i plays B in a given equilibrium and suppose at least one of them is finite. We will show that this leads to a contradiction.
   - $\tilde{q}_1 > \tilde{q}_2$: Otherwise, P1 finds it optimal to play A when reading $\tilde{q}_1$ contradicting that we are in equilibrium: ...
   - when reading $\tilde{q}_2$, P2 must weakly prefer B over A; however... $\square$

Rubinstein's electronic mail game VII

Result

Almost common knowledge can lead to very different conclusions than common knowledge.

Review questions

- Why does the presence of the announcement in the hat game change the outcome?
- What are the elments of our model of knowledge?
- How is the knowledge operator defined? Can you give an example?
- What is a self evident event?
- What are the two definitions of common knowledge?
- State Aumann's agreement theorem! What is the intuition for the result? Can you use the theorem to explain that "both players know event E" is not the same as "common knowledge"?
- What are the knowledge requirements for a Nash equilibrium (in a two player game)? How do they differ from the knowledge requirements of rationalizability?
- What is the main insight of Rubinstein's electronic mail game?

reading: OR ch. 5 (alternative: MSZ 9.1 and 9.2; *reading: MSZ 9.3-9.5)

Exercises I

1. The top of the table of the football league is 1. Team A 43, 2. Team B 41 3. Team C 40 4. Team D 36...
   The three top teams play against lower ranked teams during the next weekend. Unfortunately, you are away and cannot follow the matches. Your friend sends you the text message "Team C is now top of the table". Which of the following events do you know[1]
   - Team A lost,
   - Team A and Team B both lost,
   - Team C won and Team A lost?
2. Same as above: The top of the table of the football league is 1. Team A 43, 2. Team B 42 3. Team C 41 4. Team D 36...
   Assume A is playing against C. Assume you are only interested in the result of team A (win, draw or loss). Your friend will send you a text saying only which team is top of the table after the games. What is your information partition? In which events do you know the result of A's game?

Exercises II

3. exercise 71.2 in OR
4. Take the example on slide 19. Show that in every state (not only ω_1) the players have different posteriors for the event E. Conclude that the event "both players have different posteriors for the event E" is common knowledge. Verify that "Player i has a higher posterior than player j" is not common knowledge. (the latter can be shown in general and you are invited to give it a try; see exercise 76.1 in OR)
5. exercise 81.1 in OR: Suppose that for all $\omega \in \Omega$ all players are rational (and hence their rationality is common knowledge in every state, since any fact that is true in all states is common knowledge in every state). Show that if each player's belief in every state is derived from a common prior ρ on Ω for which $\rho(P_i(\omega)) > 0$ for all $i \in N$ and all $\omega \in \Omega$, and $a_i(\omega) = a_i(\omega')$ for each $i \in N$ and each $\omega' \in P_i(\omega)$, then $\langle(\Omega, \rho), (\mathcal{P}_i), (a_i)\rangle$,

Exercises III

where $\mathcal{P}_i$ is the partition induced by P_i , is a correlated equilibrium of G.

6. Make an exam exercise (one some topic not necessarily knowledge), i.e. a question that you could be asked in the oral exam. Send this exam question via email to me (`christoph.schottmueller@econ.ku.dk`).

[1] A win gives 3 points, a draw gives 1 point and a loss 0.